

MANGAREVA, French Polynesia

WMO: 919480

Lat: 23.130S

Lon: 134.965W

Elev: 292

StdP: 14.54

Time Zone: -9.00 (W09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	63.3	64.4	50.8	55.7	65.7	52.6	59.4	66.4	21.2	65.1	19.2	66.4	9.9	170	0.243

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.7	84.2	76.2	83.4	75.8	82.6	75.4	77.9	81.5	77.2	81.1	76.6	80.6	5.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	140.8	79.8	76.0	137.6	79.5	75.4	134.5	79.2	41.5	81.7	40.8	81.2	40.2	80.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.9	16.5	14.4	DB	61.2	85.4	1.7	0.9	60.0	86.1	59.0	86.6	58.1	87.1	56.9	87.7	(4)
(5)			WB	54.8	78.6	1.4	1.2	53.8	79.4	53.0	80.1	52.1	80.8	51.1	81.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.3	77.9	78.5	78.7	76.6	74.0	72.0	70.6	70.2	70.5	71.9	74.1	76.2		
	DBStd	3.86	1.88	1.84	1.79	2.20	2.28	2.45	2.65	2.74	2.53	2.53	2.17	2.03		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDD50	8856	866	799	891	799	745	661	639	625	614	679	724	814		
	CDD65	3385	401	379	426	349	280	211	175	162	165	214	274	349		
	CDH74	14907	2755	2874	3211	1836	722	260	138	125	158	330	775	1723		
	CDH80	1247	289	349	382	108	9	0	0	0	1	13	96			
(14) Wind	WSAvg	6.4	5.3	5.4	5.5	6.5	6.5	7.2	7.1	6.6	7.0	7.0	6.7	5.8		
(15) Precipitation	PrecAvg	70.1	6.3	5.9	6.0	5.6	5.8	5.9	5.6	5.5	4.5	5.7	6.7	7.2		
	PrecMax	102.0	19.3	15.6	20.2	19.3	15.5	12.4	9.8	19.4	10.7	13.4	19.2	22.6		
	PrecMin	38.9	0.4	0.9	0.5	0.7	1.0	1.0	1.1	0.4	0.2	0.5	1.2	0.9		
	PrecStd	14.2	4.4	3.4	4.3	3.8	3.3	3.0	2.3	3.6	2.8	3.1	4.0	4.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	85.2	85.5	83.7	81.2	79.0	78.0	77.8	78.4	79.8	81.6	83.2		
		MCWB	75.9	76.6	76.6	76.3	74.5	73.4	72.2	71.9	71.7	73.4	74.4	75.0		
	2%	DB	83.5	84.1	84.3	82.4	79.7	77.7	76.6	76.5	77.1	78.3	80.1	82.1		
		MCWB	75.5	76.2	76.3	75.6	73.8	72.7	71.5	71.1	71.2	72.3	73.5	74.6		
	5%	DB	82.6	83.2	83.3	81.3	78.5	76.7	75.5	75.5	75.9	77.2	79.1	81.1		
		MCWB	75.2	75.8	75.9	75.0	73.0	72.2	71.0	70.4	70.4	71.6	72.8	74.3		
	10%	DB	81.8	82.3	82.3	80.2	77.3	75.5	74.5	74.3	74.6	76.0	78.0	80.1		
		MCWB	74.9	75.4	75.5	74.4	72.5	71.5	70.3	69.6	69.5	71.0	72.2	73.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	78.8	78.8	78.0	76.4	74.9	73.7	73.7	73.8	75.3	76.1	77.4		
		MCDB	81.3	82.5	82.3	81.3	78.7	77.1	76.4	76.1	76.5	77.6	79.2	80.3		
	2%	WB	77.3	77.5	77.8	76.9	75.4	73.8	72.8	72.5	72.7	73.9	74.9	76.5		
		MCDB	80.9	81.8	81.7	80.8	77.8	76.3	75.5	75.1	75.5	76.6	78.1	79.6		
	5%	WB	76.5	76.9	77.1	76.1	74.6	73.2	71.9	71.5	71.7	72.9	74.2	75.7		
		MCDB	80.4	81.4	81.3	80.0	77.2	75.7	74.6	74.4	74.6	75.8	77.4	79.1		
	10%	WB	75.9	76.3	76.4	75.4	73.8	72.4	71.0	70.6	70.7	71.9	73.4	75.1		
		MCDB	79.9	80.7	80.6	79.2	76.5	74.9	73.7	73.6	73.6	74.8	76.6	78.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.1	6.7	6.0	5.5	5.4	5.6	6.4	6.4	6.5	6.5	6.8		
		MCDBR	8.0	7.9	7.8	7.3	6.6	6.5	6.9	7.4	7.9	7.7	7.6	8.0		
	5% WB	MCWBR	4.1	4.1	3.8	3.9	4.0	4.1	4.4	4.6	4.6	4.3	4.2	4.0		
		MCWBR	7.3	7.4	7.2	6.8	6.1	6.0	6.5	7.1	7.3	7.1	7.1	7.1		
(40) Clear-Sky Solar Irradiance	taub	0.366	0.369	0.362	0.366	0.358	0.351	0.346	0.347	0.358	0.359	0.362	0.369			
	taud	2.642	2.638	2.668	2.657	2.662	2.655	2.658	2.650	2.603	2.613	2.635	2.637			
	Ebn at Noon	310	305	299	285	274	270	275	286	295	305	309	310			
	Edh at Noon	32	31	29	27	25	24	25	27	30	32	32	32			
(44) All-Sky Solar Radiation	RadAvg	2136	2088	1871	1486	1196	1074	1175	1408	1697	1863	2036	2082			
	RadStd	259	180	130	146	88	78	57	74	119	157	179	187			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.72	-1.91	+0.56	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page